

City Districts 1 - 7 Linear Regression Analysis

Exploring Variable Impact on Sale Price | 2023 - 2025 Sales

Elliot Dean - Program Analyst IV
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

15 May 2026

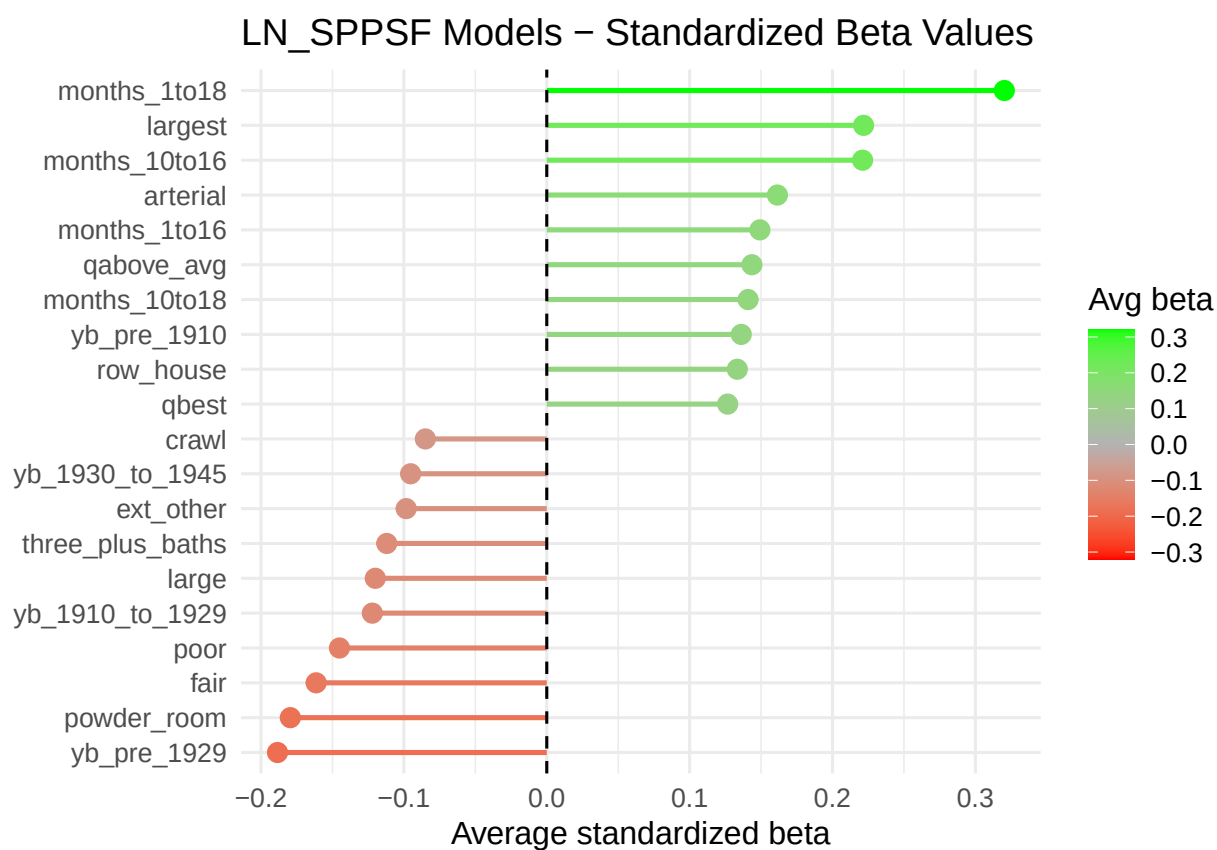


Table 1: Impact of Predictors on SPSPF

Predictor	n	Models	Avg b	Avg SE	Avg beta	Avg t	Avg sig	Min beta	Max beta
months_1to18	1	D2	0.034	0.005	0.320	6.828	0.000	0.320	0.320
largest	1	D5	0.492	0.132	0.313	3.726	0.000	0.313	0.313
qabove_avg	2	D2, D5	0.495	0.116	0.266	4.533	0.001	0.248	0.285
months_10to16	1	D5	0.055	0.016	0.221	3.479	0.001	0.221	0.221
yb_pre_1910	1	D6	0.311	0.121	0.185	2.578	0.011	0.185	0.185
arterial	2	D2, D6	0.369	0.135	0.161	2.733	0.008	0.146	0.177
months_1to16	1	D7	0.014	0.002	0.149	5.746	0.000	0.149	0.149
hot_water	1	D3	0.400	0.090	0.142	4.424	0.000	0.142	0.142
months_10to18	1	D1	0.017	0.003	0.141	6.739	0.000	0.141	0.141
qbest	1	D5	0.897	0.456	0.127	1.968	0.051	0.127	0.127
smallest	1	D4	0.249	0.062	0.126	4.005	0.000	0.126	0.126
fireplace	3	D1, D2, D4	0.131	0.037	0.123	3.542	0.001	0.079	0.168
ln_garagespaces	2	D3, D7	0.160	0.045	0.102	3.581	0.000	0.092	0.111
central_air	2	D1, D7	0.094	0.028	0.078	3.249	0.004	0.058	0.097
slab	2	D3, D6	0.088	0.179	-0.003	0.755	0.077	-0.099	0.093
yb_1946_to_1960	2	D4, D6	-0.044	0.068	-0.020	1.054	0.016	-0.158	0.119
half_duplex1story	2	D2, D4	-0.215	0.179	-0.034	-0.280	0.056	-0.113	0.046
yb_post_1990	3	D4, D5, D6	-0.223	0.191	-0.065	-2.056	0.003	-0.216	0.179
ext_brick_siding	1	D3	-0.140	0.067	-0.068	-2.104	0.036	-0.068	-0.068
qbelow_avg	2	D4, D7	-0.076	0.034	-0.073	-2.284	0.034	-0.081	-0.065
ext_other	1	D3	-0.112	0.039	-0.098	-2.845	0.005	-0.098	-0.098
yb_1930_to_1945	1	D7	-0.106	0.028	-0.103	-3.805	0.000	-0.103	-0.103
three_plus_baths	1	D6	-0.851	0.535	-0.112	-1.591	0.113	-0.112	-0.112
yb_1910_to_1929	1	D1	-0.171	0.032	-0.122	-5.334	0.000	-0.122	-0.122
poor	1	D1	-0.541	0.079	-0.145	-6.847	0.000	-0.145	-0.145
fair	1	D1	-0.211	0.028	-0.161	-7.640	0.000	-0.161	-0.161
powder_room	1	D5	-0.282	0.109	-0.179	-2.585	0.011	-0.179	-0.179
large	1	D3	-0.268	0.048	-0.186	-5.652	0.000	-0.186	-0.186
yb_pre_1929	2	D2, D7	-0.260	0.054	-0.188	-5.001	0.000	-0.216	-0.161

LN_PRICE (- ECF) Models | Standardized Beta Values

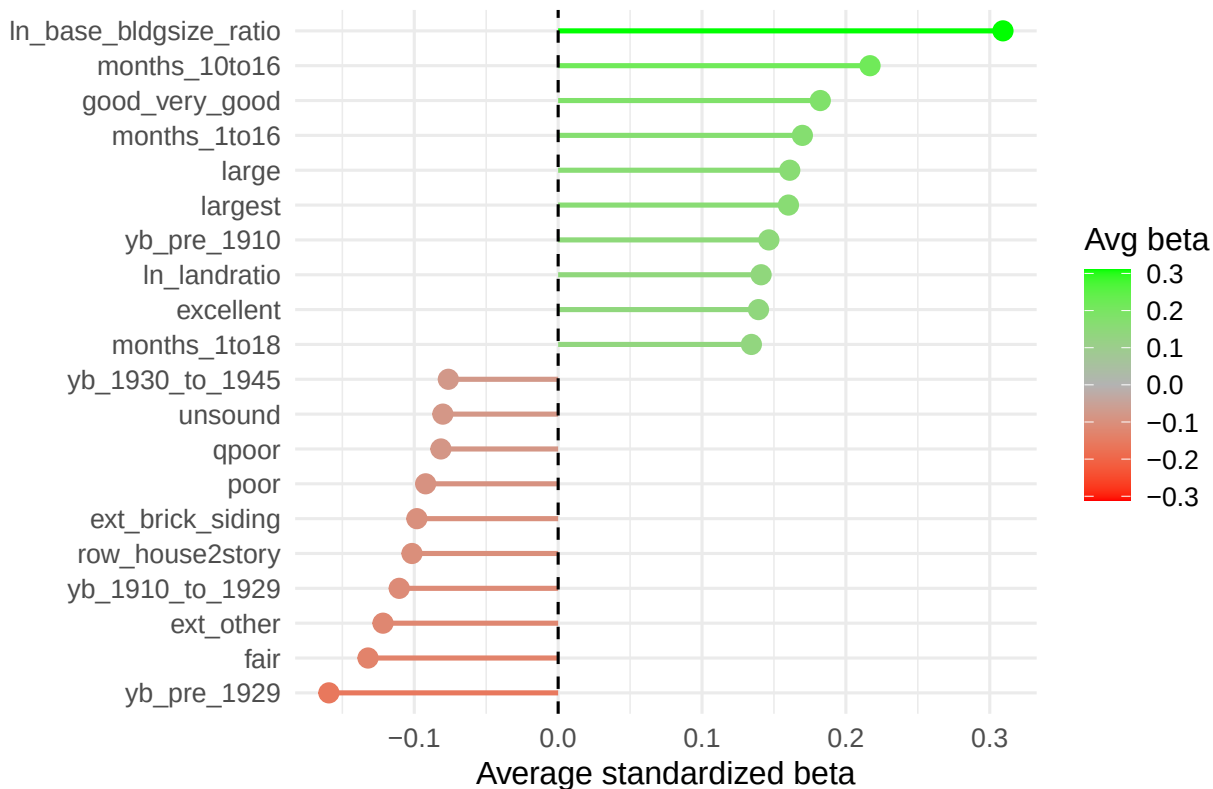


Table 2: Impact of Predictors on Saleprice (excluding ECFs)

Predictor	n	Models	Avg b	Avg SE	Avg beta	Avg t	Avg sig	Min beta	Max beta
ln_base_bldgsize_ratio	5	D1, D2, D4, D6, D7	0.705	0.110	0.309	6.960	0.000	0.225	0.407
large	1	D5	0.493	0.147	0.245	3.350	0.001	0.245	0.245
ln_landratio	2	D4, D5	0.560	0.110	0.206	5.307	0.000	0.133	0.280
good_very_good	1	D6	1.014	0.274	0.182	3.698	0.000	0.182	0.182
months_1to16	1	D7	0.017	0.002	0.170	8.083	0.000	0.170	0.170
largest	3	D1, D3, D5	0.358	0.114	0.160	3.162	0.003	0.096	0.271
yb_pre_1910	2	D3, D6	0.738	0.207	0.146	3.797	0.000	0.089	0.204
qabove_avg	1	D2	0.321	0.052	0.139	6.240	0.000	0.139	0.139
months_1to18	1	D2	0.014	0.002	0.134	7.622	0.000	0.134	0.134
months_10to18	1	D1	0.018	0.003	0.126	7.001	0.000	0.126	0.126
fireplace	1	D4	0.134	0.029	0.115	4.660	0.000	0.115	0.115
central_air	2	D3, D7	0.128	0.033	0.092	4.018	0.001	0.078	0.106
yb_1930_to_1945	1	D7	-0.081	0.026	-0.072	-3.164	0.002	-0.072	-0.072
half_duplex1story	1	D2	-0.741	0.161	-0.080	-4.597	0.000	-0.080	-0.080
poor	4	D1, D3, D4, D6	-0.392	0.099	-0.109	-4.343	0.002	-0.128	-0.062
yb_1910_to_1929	1	D1	-0.179	0.033	-0.110	-5.363	0.000	-0.110	-0.110
three_plus_baths	1	D6	-1.030	0.419	-0.118	-2.458	0.015	-0.118	-0.118
ext_other	1	D3	-0.145	0.034	-0.122	-4.261	0.000	-0.122	-0.122
yb_post_1990	1	D4	-0.372	0.071	-0.123	-5.249	0.000	-0.123	-0.123
fair	7	D1, D2, D3, D4, D5, D6, D7	-0.210	0.044	-0.132	-5.295	0.000	-0.175	-0.107
ext_brick_siding	1	D5	-0.337	0.111	-0.138	-3.046	0.003	-0.138	-0.138
yb_pre_1929	2	D2, D7	-0.243	0.035	-0.159	-6.802	0.000	-0.195	-0.124
unsound	1	D5	-1.922	0.491	-0.161	-3.915	0.000	-0.161	-0.161

LN_PRICE Models – Standardized Beta Values

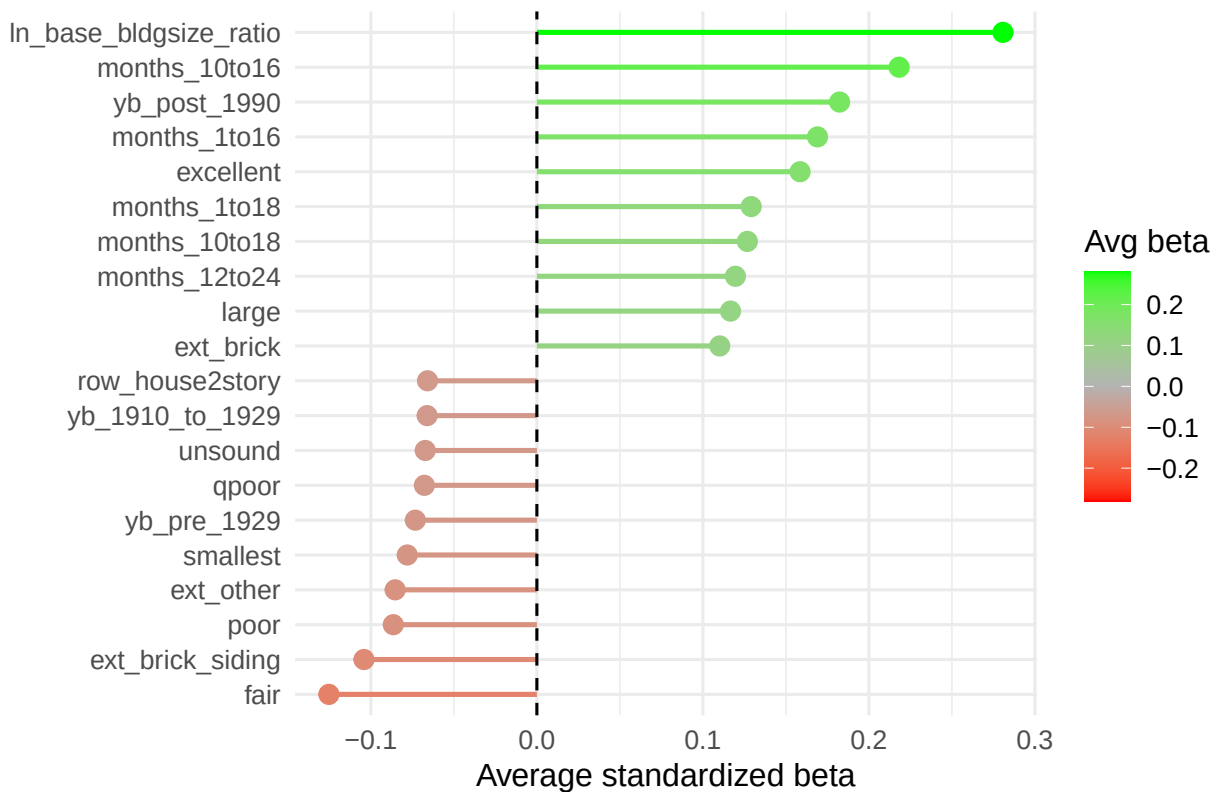


Table 3: Impact of Predictors on Saleprice

Predictor	n	Models	Avg b	Avg SE	Avg beta	Avg t	Avg sig	Min beta	Max beta
ln_base_bldgsize_ratio	5	D1, D2, D4, D5, D7	0.643	0.078	0.281	9.945	0.000	0.200	0.357
months_10to16	1	D5	0.067	0.012	0.218	5.761	0.000	0.218	0.218
yb_post_1990	1	D6	0.861	0.210	0.182	4.099	0.000	0.182	0.182
months_1to16	1	D7	0.017	0.002	0.169	8.701	0.000	0.169	0.169
ln_landratio	2	D3, D5	0.365	0.098	0.144	3.619	0.001	0.082	0.206
months_1to18	1	D2	0.014	0.002	0.129	7.897	0.000	0.129	0.129
months_10to18	1	D1	0.018	0.002	0.127	7.620	0.000	0.127	0.127
months_12to24	1	D6	0.020	0.007	0.120	2.818	0.005	0.120	0.120
largest	2	D3, D6	0.288	0.113	0.106	2.710	0.018	0.078	0.135
central_air	2	D4, D7	0.103	0.028	0.072	3.699	0.000	0.069	0.076
qabove_avg	1	D2	0.159	0.051	0.069	3.136	0.002	0.069	0.069
fireplace	1	D4	0.076	0.026	0.065	2.872	0.004	0.065	0.065
months_10to24	1	D3	0.007	0.003	0.065	2.293	0.022	0.065	0.065
good	1	D1	0.180	0.063	0.048	2.871	0.004	0.048	0.048
slab	1	D2	-0.281	0.104	-0.044	-2.693	0.007	-0.044	-0.044
yb_1910_to_1929	1	D1	-0.107	0.032	-0.066	-3.344	0.001	-0.066	-0.066
yb_pre_1929	1	D7	-0.110	0.040	-0.073	-2.760	0.006	-0.073	-0.073
smallest	1	D6	-0.217	0.115	-0.083	-1.884	0.061	-0.083	-0.083
ext_other	2	D3, D4	-0.114	0.034	-0.085	-3.380	0.010	-0.117	-0.054
half_duplex1story	1	D2	-0.802	0.149	-0.087	-5.396	0.000	-0.087	-0.087
poor	6	D1, D3, D4, D5, D6, D7	-0.327	0.092	-0.095	-3.702	0.003	-0.130	-0.051
fair	7	D1, D2, D3, D4, D5, D6, D7	-0.204	0.041	-0.125	-5.344	0.001	-0.214	-0.096
unsound	1	D5	-1.956	0.436	-0.164	-4.483	0.000	-0.164	-0.164

LN_PRICE (Inliers) Models – Standardized Beta Values

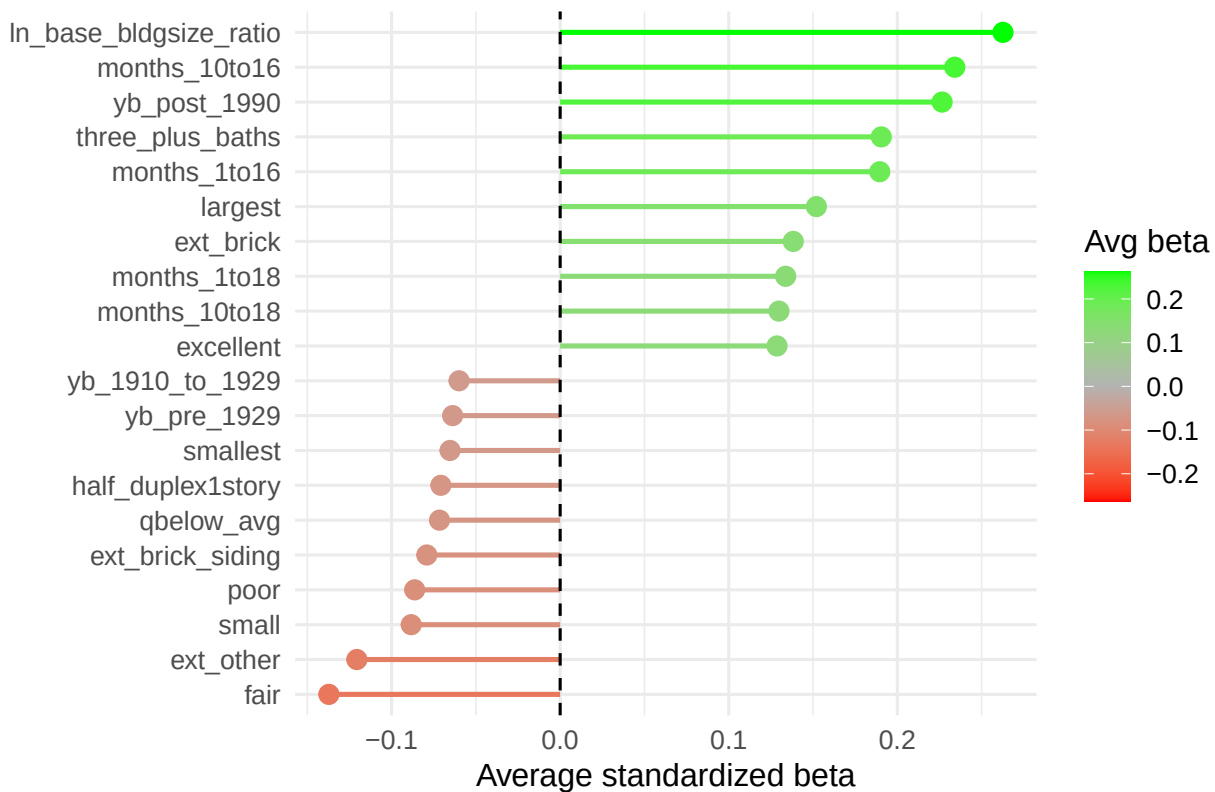


Table 4: Impact of Predictors on Saleprice

Predictor	n	Models	Avg b	Avg SE	Avg beta	Avg t	Avg sig	Min beta	Max beta
ln_base_bldgsize_ratio	5	D1, D2, D4, D6, D7	0.562	0.082	0.263	8.420	0.000	0.201	0.387
months_10to16	1	D5	0.068	0.010	0.234	7.103	0.000	0.234	0.234
yb_post_1990	1	D6	0.861	0.170	0.227	5.077	0.000	0.227	0.227
largest	2	D3, D5	0.436	0.112	0.196	3.816	0.003	0.070	0.323
months_1to16	1	D7	0.017	0.002	0.190	9.740	0.000	0.190	0.190
powder_room	1	D6	0.254	0.094	0.144	2.707	0.007	0.144	0.144
months_1to18	1	D2	0.013	0.002	0.134	8.079	0.000	0.134	0.134
months_10to18	1	D1	0.017	0.002	0.130	7.827	0.000	0.130	0.130
ln_landratio	3	D2, D3, D5	0.309	0.074	0.128	4.027	0.000	0.077	0.220
fireplace	1	D7	0.095	0.027	0.086	3.539	0.000	0.086	0.086
central_air	2	D3, D4	0.105	0.027	0.080	3.935	0.001	0.071	0.089
good	1	D1	0.260	0.062	0.071	4.233	0.000	0.071	0.071
months_15to24	1	D4	0.011	0.003	0.068	3.656	0.000	0.068	0.068
slab	1	D2	-0.410	0.122	-0.059	-3.361	0.001	-0.059	-0.059
yb_pre_1929	1	D7	-0.098	0.037	-0.064	-2.666	0.008	-0.064	-0.064
yb_1910_to_1929	1	D1	-0.113	0.030	-0.074	-3.768	0.000	-0.074	-0.074
single_family2story	1	D6	-0.117	0.068	-0.078	-1.716	0.088	-0.078	-0.078
ext_brick_siding	1	D5	-0.180	0.078	-0.079	-2.310	0.022	-0.079	-0.079
half_duplex1story	1	D2	-0.920	0.190	-0.080	-4.853	0.000	-0.080	-0.080
qbelow_avg	2	D4, D7	-0.087	0.027	-0.082	-3.298	0.004	-0.096	-0.068
poor	4	D1, D3, D5, D6	-0.404	0.116	-0.103	-4.009	0.008	-0.127	-0.088
ext_other	2	D3, D4	-0.151	0.033	-0.121	-4.767	0.000	-0.149	-0.092
fair	7	D1, D2, D3, D4, D5, D6, D7	-0.211	0.036	-0.137	-5.991	0.002	-0.269	-0.090

LN_TASP Models – Standardized Beta Values

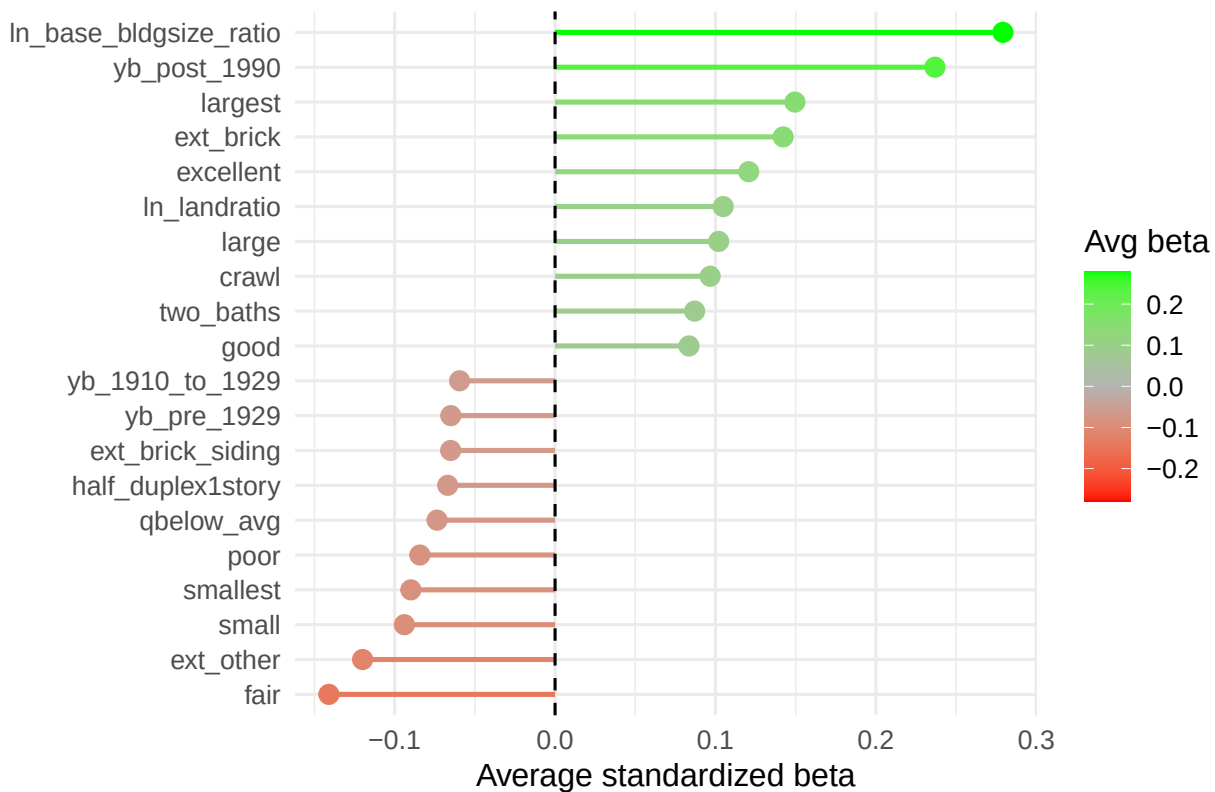


Table 5: Impact of Predictors on TASP

Predictor	n	Models	Avg b	Avg SE	Avg beta	Avg t	Avg sig	Min beta	Max beta
ln_base_bldgsize_ratio	5	D1, D2, D4, D6, D7	0.598	0.080	0.279	8.903	0.000	0.213	0.390
yb_post_1990	1	D6	0.900	0.172	0.237	5.238	0.000	0.237	0.237
largest	2	D3, D5	0.437	0.112	0.198	3.818	0.003	0.072	0.325
ext_brick	1	D6	0.234	0.071	0.142	3.277	0.001	0.142	0.142
ln_landratio	3	D2, D3, D5	0.304	0.075	0.126	3.957	0.000	0.080	0.209
fireplace	1	D7	0.089	0.027	0.082	3.342	0.001	0.082	0.082
central_air	4	D1, D3, D4, D7	0.095	0.025	0.075	3.795	0.001	0.056	0.089
good	1	D1	0.260	0.061	0.072	4.235	0.000	0.072	0.072
two_baths	1	D4	0.296	0.098	0.061	3.031	0.002	0.061	0.061
three_plus_baths	2	D5, D6	-0.210	0.290	0.042	0.833	0.017	-0.095	0.180
single_family2story	2	D2, D6	-0.018	0.046	-0.007	0.856	0.034	-0.083	0.070
yb_pre_1929	1	D7	-0.098	0.037	-0.065	-2.653	0.008	-0.065	-0.065
slab	1	D2	-0.465	0.127	-0.066	-3.662	0.000	-0.066	-0.066
arterial	1	D5	-0.197	0.098	-0.071	-2.013	0.045	-0.071	-0.071
half_duplex1story	1	D2	-0.838	0.197	-0.072	-4.250	0.000	-0.072	-0.072
yb_1910_to_1929	1	D1	-0.113	0.030	-0.075	-3.769	0.000	-0.075	-0.075
qbelow_avg	2	D4, D7	-0.087	0.027	-0.082	-3.294	0.004	-0.097	-0.068
poor	3	D1, D3, D5	-0.385	0.086	-0.106	-4.571	0.000	-0.121	-0.091
ext_other	2	D3, D4	-0.150	0.033	-0.120	-4.725	0.000	-0.148	-0.092
fair	7	D1, D2, D3, D4, D5, D6, D7	-0.214	0.036	-0.141	-6.137	0.001	-0.263	-0.091

Alternative LN_TASP Models – Standardized Beta Values

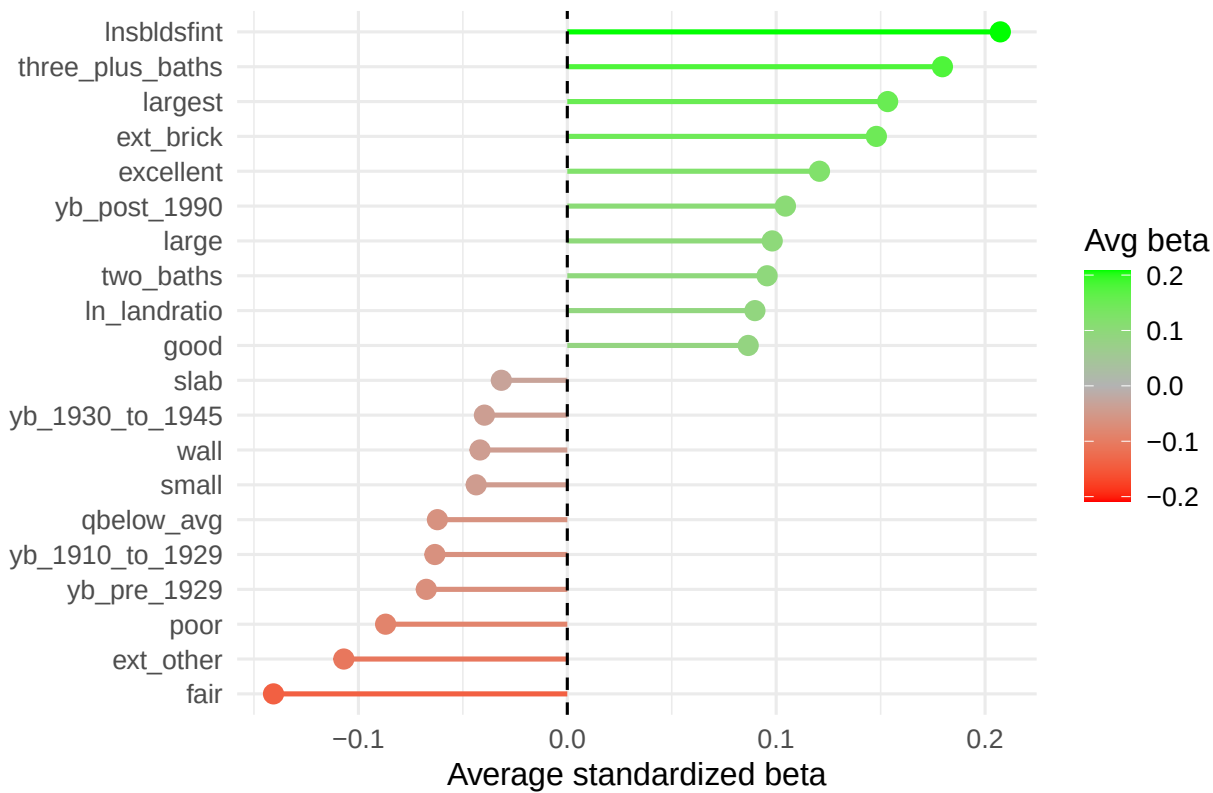


Table 6: Impact of Predictors on TASP

Predictor	n	Models	Avg b	Avg SE	Avg beta	Avg t	Avg sig	Min beta	Max beta
yb_post_1990	1	D6	0.904	0.173	0.238	5.229	0.000	0.238	0.238
largest	3	D2, D5, D6	0.423	0.106	0.217	4.189	0.001	0.114	0.325
lnsbldsfint	6	D1, D2, D3, D4, D6, D7	0.239	0.047	0.207	6.474	0.008	0.112	0.337
three_plus_baths	1	D5	0.523	0.138	0.180	3.797	0.000	0.180	0.180
ln_landratio	2	D3, D5	0.355	0.085	0.151	4.089	0.000	0.092	0.209
garage	1	D7	0.099	0.019	0.085	5.133	0.000	0.085	0.085
central_air	3	D3, D4, D7	0.104	0.026	0.077	3.998	0.000	0.063	0.090
good	1	D1	0.262	0.061	0.072	4.269	0.000	0.072	0.072
two_baths	1	D4	0.335	0.097	0.069	3.473	0.001	0.069	0.069
large	1	D1	0.077	0.029	0.068	2.680	0.007	0.068	0.068
single_family2story	2	D2, D6	-0.014	0.045	-0.007	1.397	0.029	-0.087	0.073
yb_pre_1929	2	D2, D7	-0.098	0.030	-0.068	-3.246	0.002	-0.077	-0.058
yb_1910_to_1929	1	D1	-0.106	0.030	-0.070	-3.486	0.001	-0.070	-0.070
arterial	1	D5	-0.197	0.098	-0.071	-2.013	0.045	-0.071	-0.071
unsound	1	D2	-0.753	0.128	-0.081	-5.898	0.000	-0.081	-0.081
poor	6	D1, D3, D4, D5, D6, D7	-0.355	0.099	-0.093	-4.101	0.007	-0.121	-0.063
ext_other	2	D3, D4	-0.134	0.033	-0.107	-4.160	0.000	-0.131	-0.083
fair	7	D1, D2, D3, D4, D5, D6, D7	-0.215	0.035	-0.141	-6.631	0.002	-0.263	-0.103